Introduction to Computer Science Fall 2026
Homework of Week 6

Due: November 1, 2026 at 23:58:59
No Copy-and-Paste. No Late Submission.

Instructions. Please write your answers to each problem using LaTeX and submit the compiled PDF file. If you are unfamiliar with LaTeX, you may refer to https://latex-tutorial.com/ or other tutorials for help.

If you find a problem difficult to understand or solve, you may use external resources, including search engines (such as Google), online references (such as Wikipedia), and AI tools (such as DeepSeek), to study relevant concepts and assist your learning. However, you are responsible for understanding your submitted solution, and **please make sure every word in your submission is written by yourself to avoid plagiarism**.

If you use any AI tool while working on the assignment, please include a short disclosure at the end of your submission stating which tool you used and briefly describing how you used it. Undisclosed use of AI tools may be subject to point deductions.

1 Bloch Sphere (2 pts)

A single-qubit pure state can be represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Note that a global phase does not affect measurement outcomes, so for a scalar z with $|z| = 1$, $|\psi\rangle$ is equivalent with $z|\psi\rangle$. Therefore, any single-qubit pure state can be written in the form

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

where $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$. Its corresponding Bloch vector is

$$\vec{r} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$$

(Fig. 1).

a) (1 pt) Determine the Bloch vector for the following quantum state: $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$.

b) (1 pt) Determine the Bloch vectors for the quantum states $H|\psi\rangle$ and $T|\psi\rangle$, where

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad T = \begin{pmatrix}1 & 0\\ 0 & e^{i\pi/4}\end{pmatrix}.$$

2 Evolution (2 pts)

An isolated quantum system evolves according to the Schrödinger equation:

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle,$$

where H is the Hamiltonian.

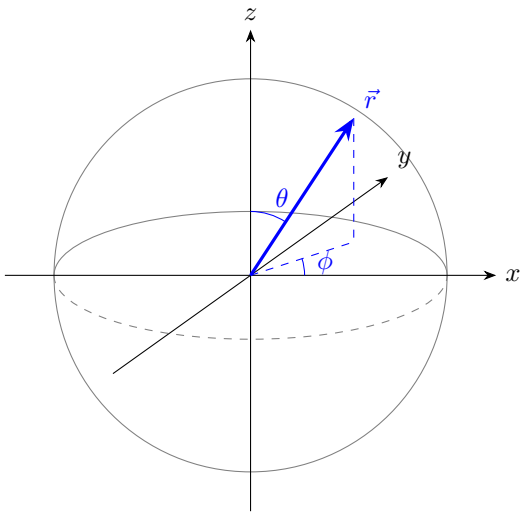


Figure 1: Bloch sphere representation of a single-qubit pure state. Any single-qubit pure state can be represented by a Bloch vector $\vec{r}$ on the unit sphere.

Consider a single-qubit system with time-invariant

$$H = \hbar\omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad |\psi(0)\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Here ω is an arbitrary real number.

a) (1 pt) Compute the quantum state $|\psi(t)\rangle$ at time t.

b) (1 pt) Compute the probability distributions of measuring $|\psi(0)\rangle$ and $|\psi(t)\rangle$ in the X basis, where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

3 Entanglement (6 pts)

When we measure a qubit, the outcome appears random. But for many years, physicists wondered whether the outcome might actually be determined in advance, and we simply did not know it.

Experimental Setup. Alice and Bob are given many identically prepared pairs of qubits. For each pair, Alice holds the first qubit and Bob holds the second. Each pair is used only once.

Alice can choose between two measurement bases, A_0 and A_1, while Bob can choose between B_0 and B_1. Each basis consists of two orthogonal states, corresponding to outcomes $+1$ and -1.

Let $E(A_i, B_j)$ denote the average product of their measurement outcomes when Alice uses A_i and Bob uses B_j. Define the CHSH quantity as

$$S = E(A_0, B_0) + E(A_0, B_1) + E(A_1, B_0) - E(A_1, B_1).$$

Part I: Predetermined Outcomes.

Suppose that all possible measurement outcomes are determined in advance. For each pair, denote them by $a_0, a_1, b_0, b_1 \in \{-1, +1\}$. Each outcome depends only on the local choice of basis, not on the other person's choice. These predetermined outcomes may differ between pairs.

a) (1 pt) Compute the maximum possible value of

$$|a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1|,$$

where $a_0, a_1, b_0, b_1 \in \{-1, +1\}$.

b) (1 pt) Suppose the predetermined outcomes may differ between pairs. Compute the maximum possible value of $|S|$.

Part II: Quantum Entanglement.

Now suppose that each pair is prepared in the entangled state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Alice and Bob use the following measurement bases:

$$\begin{aligned}
A_0 &: \{|0\rangle,\ |1\rangle\}, \\
A_1 &: \{|+\rangle,\ |-\rangle\}, \\
B_0 &: \left\{ \begin{array}{l} \cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle, \\ \sin\frac{\pi}{8}|0\rangle - \cos\frac{\pi}{8}|1\rangle \end{array} \right\}, \\
B_1 &: \left\{ \begin{array}{l} \cos\frac{\pi}{8}|0\rangle - \sin\frac{\pi}{8}|1\rangle, \\ \sin\frac{\pi}{8}|0\rangle + \cos\frac{\pi}{8}|1\rangle \end{array} \right\}.
\end{aligned}$$

For each basis, the first state corresponds to outcome $+1$ and the second to outcome -1.

c) (2 pts) Compute the four correlations $E(A_0, B_0)$, $E(A_0, B_1)$, $E(A_1, B_0)$, and $E(A_1, B_1)$.

d) (2 pts) Compute S using your results from (c). Is it larger than the result of (b)?

Declaration of AI Use

Please disclose any use of generative AI tools in completing this homework. For each tool used, briefly state what you used it for. You remain responsible for the correctness and originality of your submission. If you did not use any generative AI tools, remove this section.